Postmortem: card payment outage on 2026-03-12

Document: INC-301

Date: 2026-03-14

Services: payments-api, ledger, card-gateway, notifications

Summary

SEV1 postmortem of the March 12, 2026 outage affecting card payment processing. Change CHG-1042 reduced the ledger connection pool max_connections from 200 to 20, causing ledger connection pool exhaustion. This cascaded across payments-api, card-gateway, and notifications services. Incident commander Elena Petrova. Report author Maya Chen, Payments Platform team lead.

Impact

Card payments failed for 1,240 merchants over 56 minutes. payments-api returned errors to merchants attempting to create payments. card-gateway experienced timeouts while waiting for ledger writes to complete. notifications service delayed 18,400 webhooks to merchants about payment events. See STS-301 for status updates during the incident.

Timeline

- 14:02 UTC: impact began at 14:02 UTC; ledger connection pool exhaustion detected
- 14:15 UTC: Incident declared SEV1
- 14:41 UTC: CHG-1042 rolled back
- 14:58 UTC: recovered at 14:58 UTC; all payments processing normally

Root cause

CHG-1042 was approved by Ravi Patel and deployed to reduce ledger connection pool size as a cost optimization. The change set max_connections from 200 to 20 without accounting for peak payment throughput. Under normal load, the reduced pool became exhausted, causing connection timeouts across all services that depend on ledger writes. See INC-302 for card-gateway team's preliminary report on authorization timeouts.

Resolution

CHG-1042 was rolled back at 14:41 UTC, restoring the connection pool to its previous size. CHG-1047 subsequently restored the ledger pool size and added a configuration check to prevent similar reductions in the future.

Follow-up actions

- Review RB-03 for ledger connection pool saturation procedures
- Implement capacity planning review for all service dependencies
- Add monitoring alerts for connection pool saturation across all services